

Miniproject: General Instructions

Computational Neuroscience: Neuronal Dynamics (NX-465)

Spring semester 2026

General Instructions – Mini-Projects

As part of the Neuronal Dynamics course, you will work on one of the two proposed mini-projects:

1. *Visual Surround Suppression using Excitation-Inhibition Circuits*
2. *Spatial Navigation using Grid Cells*

Grading The mini-project is **mandatory** and counts towards 30% of the final grade. You will be graded on the results you obtained and your answers to the questions, and a fraction of the grading is also dedicated to the quality and clarity of the report. The mini-project grading totals 60 points, 10 of which are dedicated to the quality and clarity of the report. The remaining 50 points are dedicated to the results. The bonus exercises are not mandatory (you can reach the maximum grade without solving them), but can give you a few extra points.

Teams You will have to work in teams of two students. To find a project partner if you don't have one yet, you can use the EdStem forum (<https://edstem.org/eu/courses/3079/discussion/227739>).

Tip: Dividing the projects into two parts such that each partner works independently on its own might not be the best strategy. The projects are very sequential and build upon the understanding of the preceding exercise, we therefore encourage exchange of knowledge between the team members. Let us know, and realise in time, if your project partner is going to drop the course such that we can provide extra help to finish the project.

Final report Document all your solutions in a short project report (one per group; max. 7 pages). This report will be the basis of your group grade and should contain:

- a brief description of the topic you are dealing with in your own words (between 5 to 10 lines),
- the answers to the questions in the text, including the figure of each simulation requested,
- a short conclusion (between 5 to 10 lines).
- your answer to one final question
- clarification on your approach (specified below)

Please explain within half a page, how you approached the different simulation tasks in the miniproject. Did you code from scratch, or get inspired by existing programs (if so which ones, which bits did you copy from, which did you rewrite), or used a LLM (if so, how did you structure the prompts for the interaction with the LLM). You should write in the style: I did/we did/the team of x+y did ... and explain step by step, figure by figure how you proceeded.

Write concisely in complete English sentences and present figures carefully (axis labels, legends etc.). Make sure to put all the figures and simulation results requested in your report. Simulation results that appear only in the code will not be graded.

Python implementation The mini-projects contain both theory questions and coding questions. The latter are intended to be solved without using any specific library (other than the usual `numpy` and `matplotlib`, etc.).

The python files with your code have to be submitted together with the final report. It will not be evaluated, but it will be used for fraud detection, so make sure we can follow how you generated your figures.

Submission You will have to submit your Python code and your project report in a single zip file via the moodle page. The file name should have the structure `MPX_NameMember1_NameMember2.zip`, where X is the miniproject number (either 1 or 2). The zip file should contain:

- a single pdf file with you project report:
`Report_MPX_NameMember1_NameMember2.pdf`
- and a folder with your source code:
`Code_MPX_NameMember1_NameMember2/...`

If you work in teams both of you should submit the same zip file. On moodle, make sure you press the submit button before the deadline.

Deadline The deadlines to submit your code and report are:

- May 26th 2026, at 11.55 pm (fraud-detection slots: May 27th and 28th, 2026);
- June 1st 2026, at 11.55 pm (fraud-detection slots: June 2nd and 3rd, 2026).

The early deadline allows you to do the fraud detection before the exam preparation period, which you might find convenient.

Regulations and fraud detection After the submission, we will perform a fraud detection interview which consists of a few questions about your report and your code. You will need to attend **in-person**.

The mini-project is graded and the same rules as for exams apply (“Ordonnance sur le contrôle des études à l’EPFL”). In short: EPFL takes any form of cheating and plagiarism very serious. The mini-project has to be your own work. You are not allowed to share code or answers across teams. However you can discuss ideas between teams and ask questions on the Ed forum. If you split the work between the two team members, or if you use a large language model, both team members have to understand all parts of the mini-project and have a good understanding of the submitted code. Note: The goal of this meeting is fraud detection; it is neither an exam nor a presentation.